

KineST: A Kinematics-guided Spatiotemporal State Space Model for Human Motion Tracking from Sparse Signals

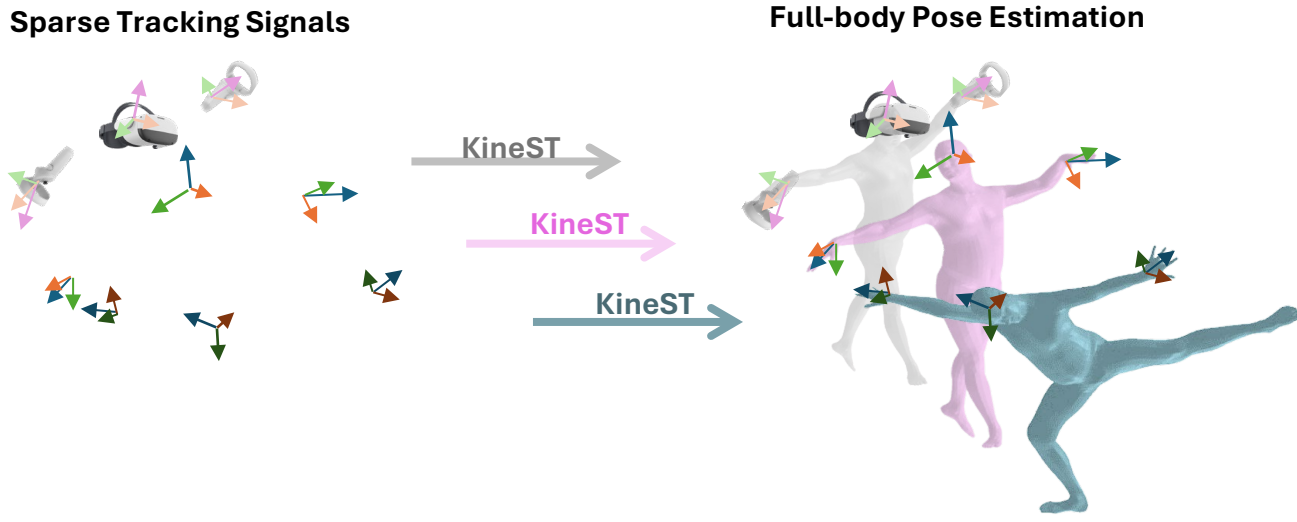
If you're interested please
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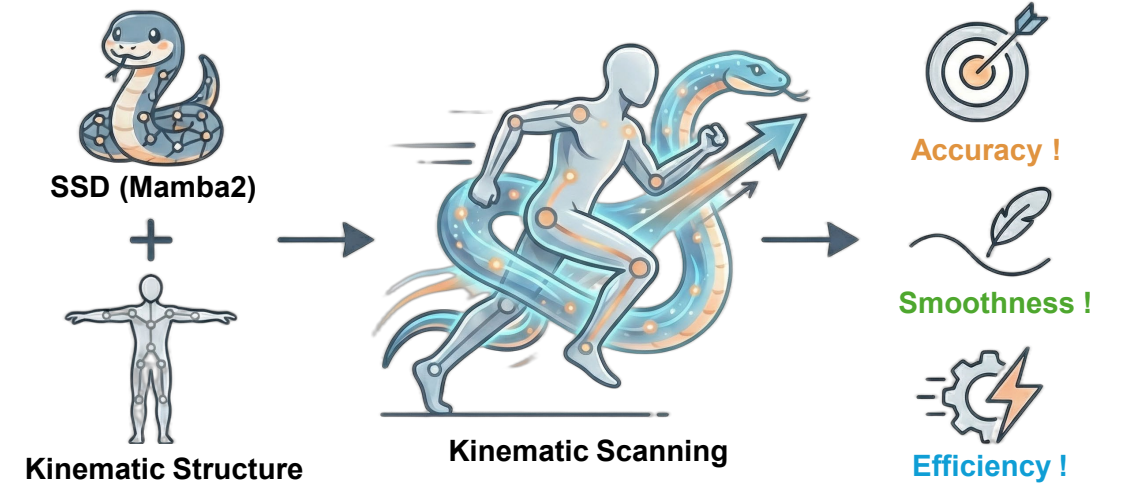
Looking for PhD
positions !!!
(26 Fall)



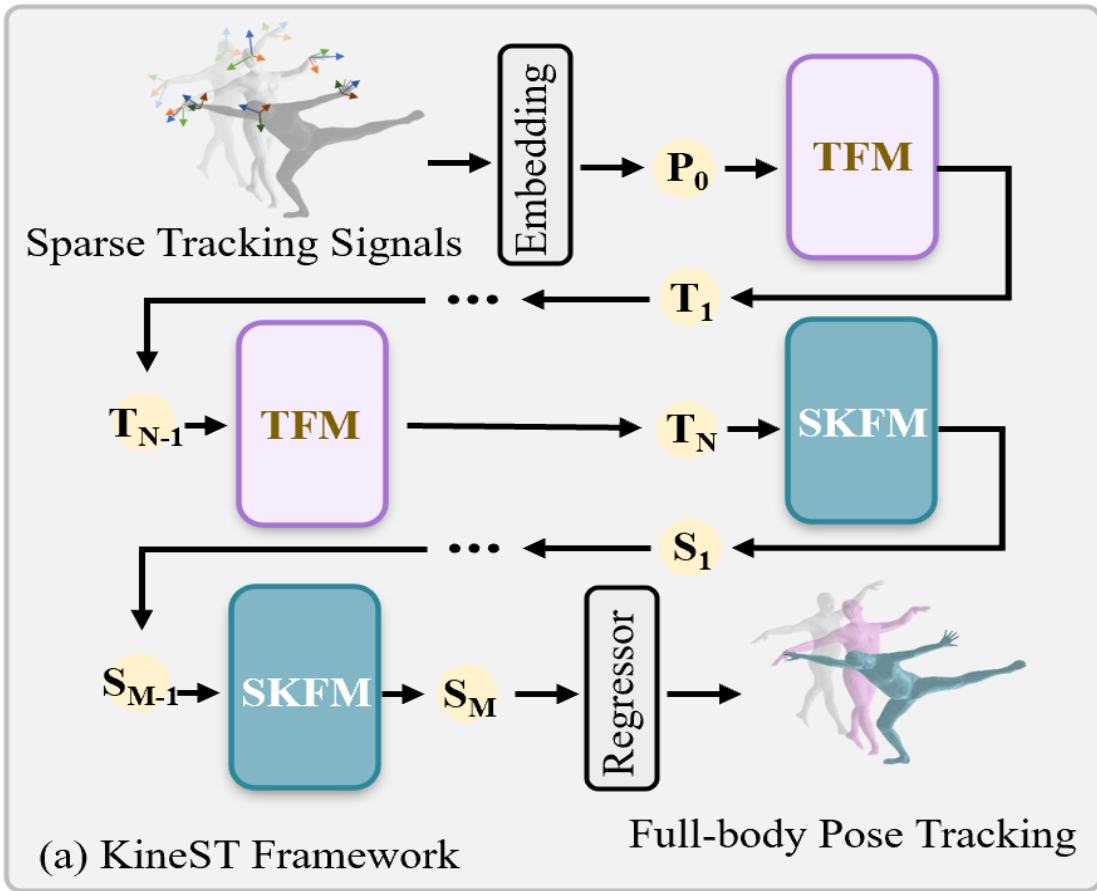
Task



Motivation

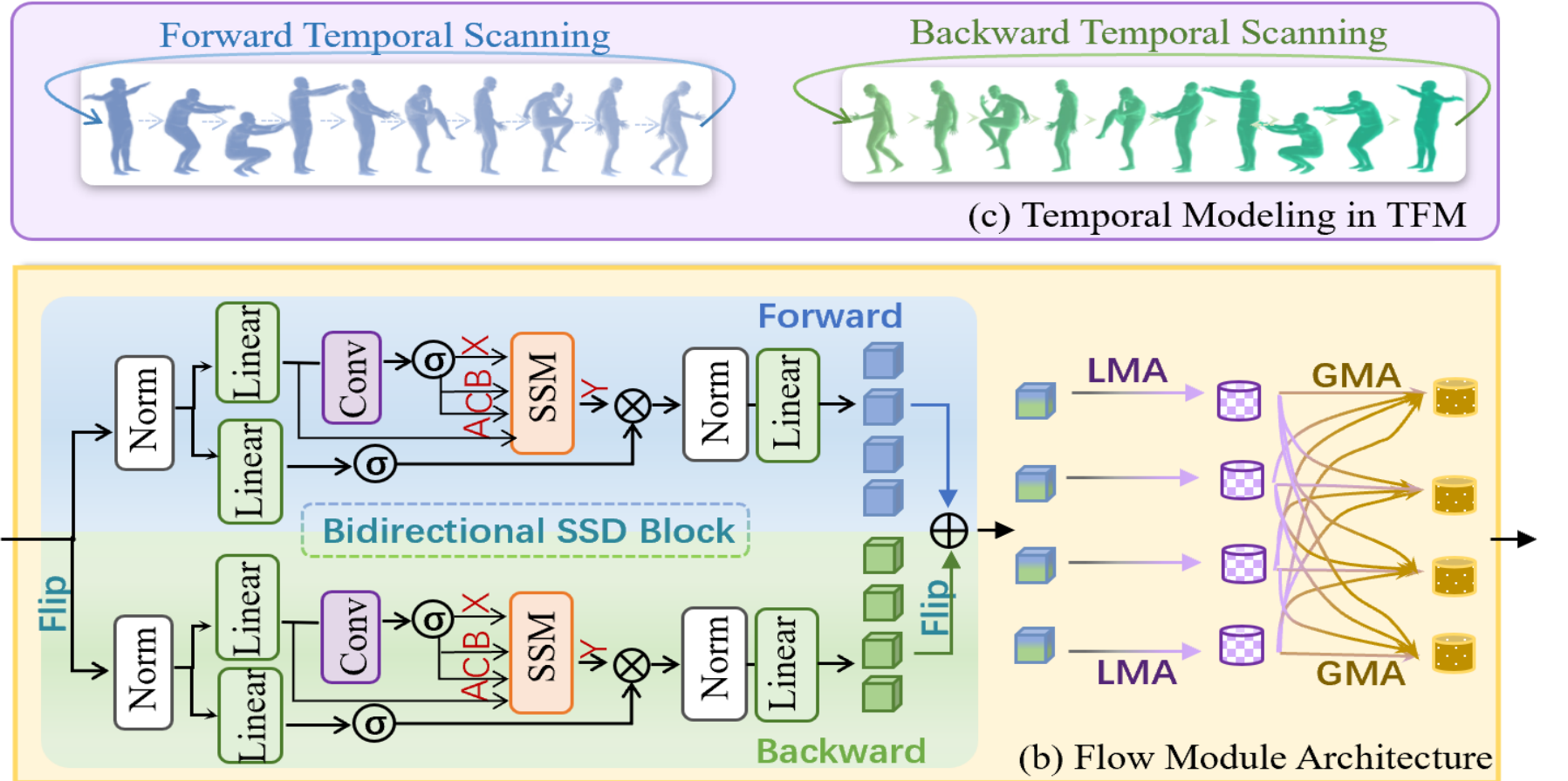


Method



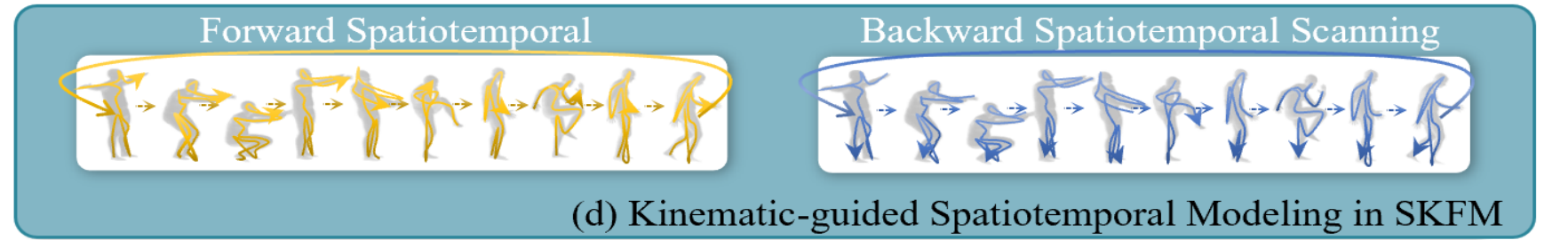
(a) KineST Framework

Full-body Pose Tracking



(c) Temporal Modeling in TFM

(b) Flow Module Architecture



(d) Kinematic-guided Spatiotemporal Modeling in SKFM

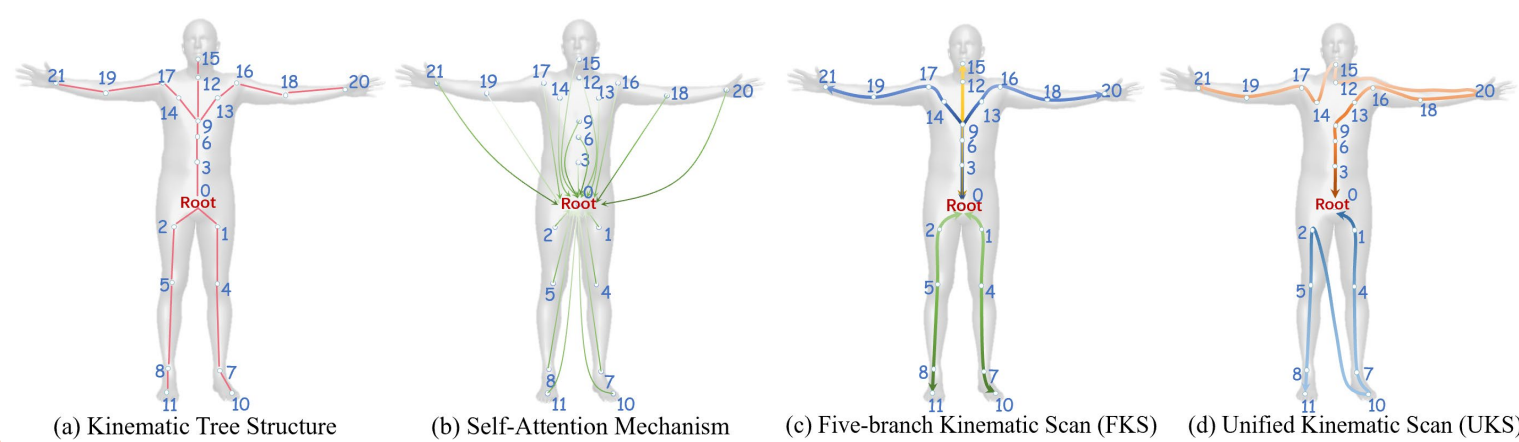
Spatiotemporal Mixing Mechanism

Input: Final temporal motion features $T_N \in \mathbb{R}^{L \times E}$
Output: Final spatiotemporal motion features $S_M \in \mathbb{R}^{L \times E}$

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1: for i in M do
2:   if i = 0 then  $S_i \leftarrow \text{Linear}(T_N)$ 
3:   else  $S_i \leftarrow \text{Linear}(S_{i-1})$ 
4:   end if
5:    $S'_i \leftarrow \text{rearrange}(S_i, L F \rightarrow L (J D))$ 
6:    $S''_i, S'''_i \leftarrow \text{KTSS}(S'_i)$ 
7:    $S'_i \leftarrow \text{rearrange}(S'_i, L J F D \rightarrow (L J_f) D)$ 
8:    $S''_i \leftarrow \text{rearrange}(S''_i, L J_b D \rightarrow (L J_b) D)$ 
9:    $S'_i, S''_i \leftarrow \text{Bi-SSD}(S'_i, S''_i)$ 
10:   $S_{i+1} \leftarrow \text{GMA}(\text{LMA}(\text{Linear}(S'_i + S''_i)))$ 
11: end for
12: return  $S_M$ 
    
```

Kinematic Tree Scanning Strategy



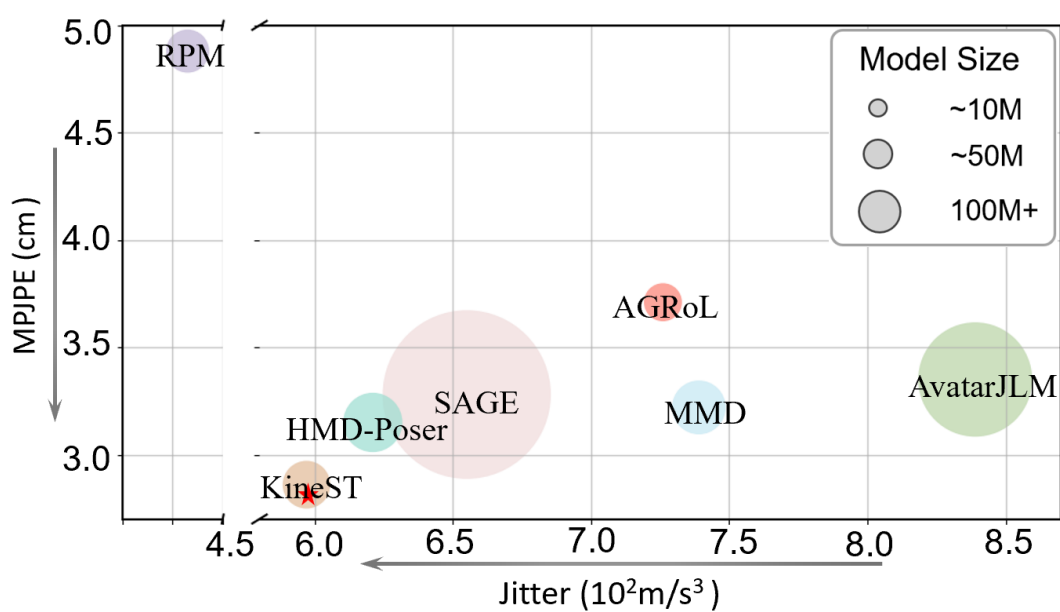
Geometric Angular Velocity Loss

$$\theta_V = \arccos\left(\frac{\text{Tr}(V) - 1}{2}\right),$$

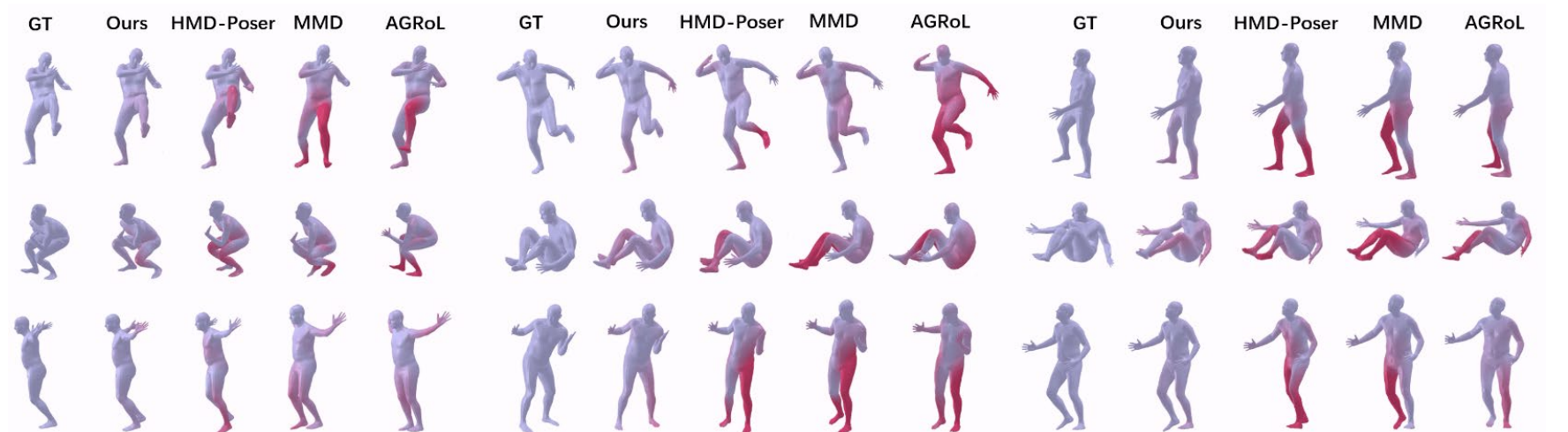
$$\log V = \theta_V \cdot \frac{1}{2 \sin \theta_V} \begin{bmatrix} V_{32} - V_{23} \\ V_{13} - V_{31} \\ V_{21} - V_{12} \end{bmatrix},$$

$$\mathcal{L}_{\text{angvel}}^{\text{geo}} = \sum_{t=1}^{T-1} \left\| \log(V_t) - \log(\hat{V}_t) \right\|_1.$$

Results



Method	MPJRE↓	MPJPE↓	MPJVE↓	Hand PE↓	Upper PE↓	Lower PE↓	Root PE↓	Jitter↓	Param.
AvatarPoser (Jiang et al. 2022a)	3.08	4.18	27.70	2.12	1.81	7.59	3.34	14.49	4M
AvatarJLM (Zheng et al. 2023)	2.90	3.35	20.79	1.24	1.72	6.20	2.94	8.39	63M
AGRoL (Du et al. 2023)	2.66	3.71	18.59	1.31	1.55	6.84	3.36	7.26	7M
SAGE (Feng et al. 2024)	2.53	3.28	20.62	1.18	1.39	6.01	2.95	6.55	137M
HMD-Poser* (Dai et al. 2024a)	2.32	<u>3.15</u>	18.15	1.35	1.34	<u>5.76</u>	<u>2.76</u>	6.21	17M
MMD (Dong et al. 2024)	<u>2.31</u>	3.22	<u>17.88</u>	0.94	<u>1.29</u>	6.01	2.88	7.39	14M
RPM* (Barquero et al. 2025)	3.69	4.88	21.99	5.90	2.94	7.69	4.01	4.43	9M
KineST (Ours)	2.25	2.86	15.26	<u>1.04</u>	1.24	5.20	2.65	<u>5.97</u>	11M
GT	0	0	0	0	0	0	0	4.00	—



Scan It !

